

Grade 6 Accelerated
Unit 1
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

Find the greatest common factor (GCF) of each pair of numbers.

1. 20 and 32

20: 1, 2, 4, 5, 10, 20
32: 1, 2, 4, 8, 16, 32

GCF = 4

2. 60 and 486

60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60
486: 1, 2, 3, 6, 9, 18, 27, 54, 81, 162, 243, 486

GCF = 6

3. 13 and 9

13: 1, 13
9: 1, 3, 9

GCF = 1

4. 392 and 616

392: 1, 2, 4, 7, 8, 14, 28, 49, 56, 98, 196, 392
616: 1, 2, 4, 7, 8, 11, 14, 22, 28, 44, 56, 77, 88, 154, 308, 616

GCF = 56

5. Dr. Harris has 90 sixth-grade students and 75 seventh-grade students in her chorus class. She wants to make groups of performers, with the same amount of sixth- and seventh-grade students in each group; keeping in mind that she wants to form as many groups as possible. What is the largest number of groups that could be formed?

90: 1, 2, 3, 5, 6, 9, 10, 15, 18, 30, 45, 90
75: 1, 3, 5, 15, 25, 75

The largest group can have 15 students.

Rewrite each fraction in simplest form using the GCF of the numerator and denominator.

6. $\frac{30}{55}$

7. $\frac{17}{68}$

$$\frac{30}{55} = \frac{\cancel{5} \times 6}{\cancel{5} \times 11} = \frac{6}{11}$$

$$\frac{17}{68} = \frac{1 \times \cancel{17}}{4 \times \cancel{17}} = \frac{1}{4}$$

8. $\frac{306}{630}$

9. $\frac{72}{244}$

$$\frac{306}{630} = \frac{\cancel{18} \times 17}{\cancel{18} \times 35} = \frac{17}{35}$$

$$\frac{72}{244} = \frac{18 \times 4}{61 \times 4} = \frac{18}{61}$$

10. Marguerite wanted to rewrite the fraction $\frac{760}{800}$ in simplest form. She said, "The GCF of 760 and 800 is 10. Since $760 \div 10 = 76$ and $800 \div 10 = 80$, I can rewrite $\frac{760}{800}$ in simplest form as $\frac{76}{80}$." Write a note to Marguerite explaining why she is incorrect and what she can do to correct her mistake.

It can be simplified even more.

$$\frac{76}{80} = \frac{19 \times 4}{20 \times 4} = \frac{19}{20}$$